

第八章作业

8.31

$t < 0$ 时, 电感短路, 电容开路, 则

$$V_C(0^-) = 50 \times \frac{40}{40+10} = 40V, \quad V_C(0^+) = V_C(0^-) = 40V$$

电感电流不能突变, $i_L(0^-) = 1A$, 所以 $i_L(0^+) = 1A$,

在 0^+ 时电感上端节点列 KCL, $1 + \frac{V_C(0^+) - 40}{40} - 2 = 0$

可解得 $V_C(0^+) = 80V$

$$\text{即 } V_C(0^+) = 40V \quad V_L(0^+) = 80V$$

8.32

① $t < 0$ 时, 电感短路, 电容开路, $i_L(0^-) = 0$, 对左节点列 KCL: $\frac{V_A}{4} + 3 = 0$,

则 $V_A = -12V$, 对右节点列 KCL: $\frac{V_B}{2} - 3 = 0$, 则 $V_B = 6V$

则 $V_L(0^-) = V_A - V_B = -12 - 6 = -18V$, $i_L(0^+) = 0$

② $t > 0$ 时, 电流源变为 0, 相当于开路, 电压源为 $75V$, 设中间支路电流为 i , 方向从左到右,

$i = C \frac{dV}{dt}$, 由 $C = 0.04F$, 则 $i = 0.04V'$

左侧电压为 V_A , 右侧电压为 V_B ,

左节点 KCL: $\frac{V_A - 75}{4} + i = 0$, $V_A = 75 - 4i$

右节点 KCL: $\frac{V_B}{2} - i = 0$, $V_B = 2i$

$V_A - V_B = V_L + V$, $V_L = L \frac{di}{dt}$, $L = 1H$, 则 $V_L = \frac{di}{dt}$,

即 $V_A - V_B = \frac{di}{dt} + V$, 即 $75 - 4i - 2i = \frac{di}{dt} + V$, 又由 $i = 0.04V'$

则 $0.04V'' + 6(0.04V') + V = 75$, 化简得

$$V'' + 6V' + 25V = 1875, \quad V_p = 75$$

特征方程: $s^2 + 6s + 25 = 0 \Rightarrow s = -3 \pm j4$

则通解: $V(t) = 75 + e^{-3t}(A \cos 4t + B \sin 4t)$,

代入初值: $V(0^+) = -18$, $75 + A = -18$ 即 $A = -93$

又 $V'(0^+) = -3A + 4B = 0$, 代入 A , 可得 $B = -\frac{279}{4} = -69.75$

故 $V(t) = 75 - e^{-3t}(93 \cos 4t + 69.75 \sin 4t)V, \quad t > 0$

8.47

$t = 0^-$ 时, $i_L(0^-) = \frac{6 \times 5}{10+5} = 2A$, 且 $V_C(0^-) = 0$,

$t > 0$ 时, 电路变为并联 RLC 电路

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 5 \times 0.01} = 10, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.01}} = 10$$

由于 $\alpha = \omega_0$, 为临界阻尼响应, 则电感电流为

$i(t) = I_s + (A + Bt)e^{-10t}$, $I_s = 6A$, 由 $i(0) = 2 = 6 + A$, 则 $A = -4$

$V_C = L \frac{di}{dt} = Be^{-10t} - 10(A + Bt)e^{-10t}$, 由 $V_C(0) = 0$, 则 $B - 10A = 0$.

可得 $B = -40$, 则 $V_C(t) = [-40 - 10(-4 - 40t)]e^{-10t} = 400te^{-10t}V, \quad t > 0$

$$\text{则 } V_C(t) = \begin{cases} 0, & t < 0. \\ 400te^{-10t}V, & t \geq 0 \end{cases}$$

8.48

$$t=0^- \text{ 时, } i(0) = \frac{-b}{j} = -2A, \quad v(0) = 2 \times 1 = 2V$$

$t > 0$ 时, 为串联RLC电路,

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 1 \times 0.25} = 2, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.15}} = 2$$

由 $\alpha = \omega_0$, 其为临界阻尼,

$$\text{则) } i(t) = [A + Bt]e^{-2t}, \quad i(0) = -2 = A,$$

$$v = L \frac{di}{dt} = B e^{-2t} + [-2(-2 + Bt)e^{-2t}], \quad v(0) = 2 = B + 4, \quad B = -2$$

因此

$$i(t) = [(-2 - 2t)e^{-2t}]A, \quad v(t) = [2 + 4te^{-2t}]V$$

8.53

$$t < 0 \text{ 时, } i(0^-) = 0, \quad v_L(0^-) = 120V,$$

$$t > 0 \text{ 时, } \frac{120 - v}{R} = C \frac{dv}{dt} + i \quad \text{EP} \quad 120 = v + RC \frac{dv}{dt} + iR.$$

$$\text{又由 } v_L = v = L \frac{di}{dt}, \text{ 则}$$

$$120 = L \frac{di}{dt} + RCL \frac{d^2i}{dt^2} + iR \quad \text{EP} \quad 120 = \frac{1}{4} \frac{di}{dt} + 80 \times \frac{1}{4} \times 10 \times 10^{-3} \frac{d^2i}{dt^2} + 80i$$

也即

$$\frac{d^2i}{dt^2} + 0.125 \frac{di}{dt} + 400i = 600$$